Problem Set 4: Question 3

Econ 508: Applied Econometrics

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November 20, 2013

Motivation

- In this question we are concerned on how much changes in productivity during different times of the Phuzicists careers would affect their lifetime earnings.
- To solve this question we adopt the following strategy
  - First, we estimate a salary model, to understand how productivity affects wages
  - Then we simulate productivity increase at different stages of the researcher productive life.

First Step

Our salary model is the same as suggested by Divide and Koenker (1991)

$$log\left(\frac{s_{it}}{s_{it-1}}\right) = \beta_0 + \beta_1 log\left(\frac{Y_{it}}{Y_{it-1}}\right) + \beta_2 ru_{it} + \beta_3 Sex_i + \alpha_i + u_{it}$$

- Where the dependent variable is the difference between the logarithm of salary in time$t$ and the logarithm of salary in time $t-1$.
- Whereas ru_{it} is an indicator of employment in research university of individual i in year t.

First Step: Results

Table 3: Estimates for Phuzicist salaries

	Pooled OLS	Between	Within	GLS	HTIV
Productivity	0.052***	0.055***	0.051***	0.052***	0.052***
	(0.002)	(0.003)	(0.002)	(0.002)	(0.002)
=1 if Female	-0.005***	-0.006***		-0.006***	-0.006***
	(0.001)	(0.001)		(0.001)	(0.002)
=1 if at Research Institution	0.008***	0.009***	0.004***	0.008***	0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
θ					0.017***
					(0.001)
Constant	0.015***	0.016***	0.014***	0.015***	
	(0.001)	(0.001)	(0.001)	(0.001)	
Observations	5945	5945	5945	5945	5945
Number of phuzicists		497	497	497	497
Joint Significance Test	516.3	129.6	528.3	1438.1	65064.2
p-value	0.0000	0.0000	0.0000	0.0000	0.0000
Hausman Test			25.9		
p-value			0.0000		

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

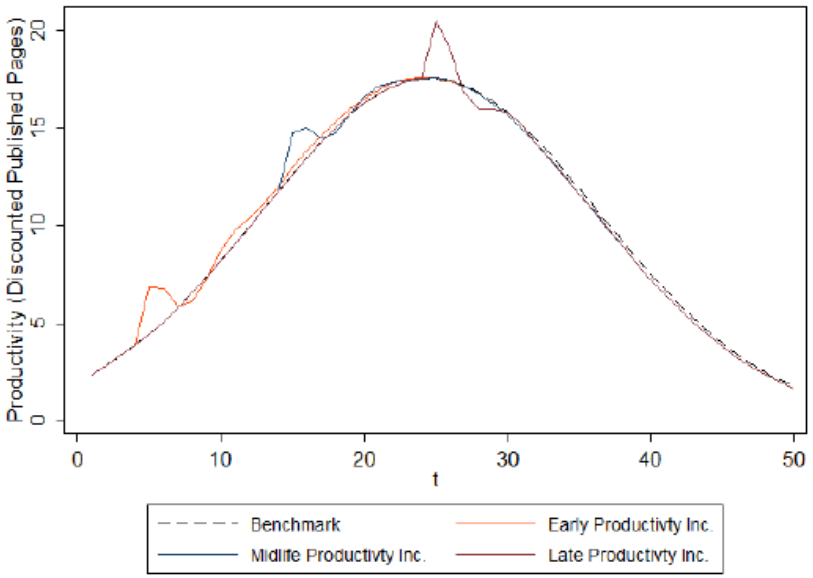
Second Step: Simulations

- Once we have a model linking salaries and productivity, we conduct our simulations.
- We consider a productivity innovation as an additional published research paper (3 pages).
- And look at 4 Scenarios for men and women

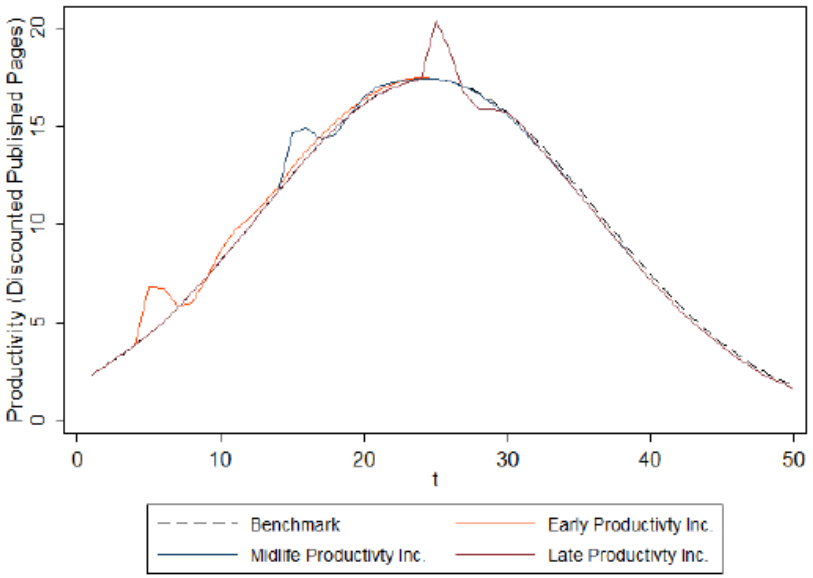
  1. Benchmark: no innovation
  2. Early productivity change: innovation in 5th year of career
  3. Mid-life: innovation in15th year of career
  4. Late productivity change: innovation in 25th year of career T

Note: all simulatons are based on HTIV estimates, and we transform all the variables consequently.

Second Step: Productivity Simulations



Men

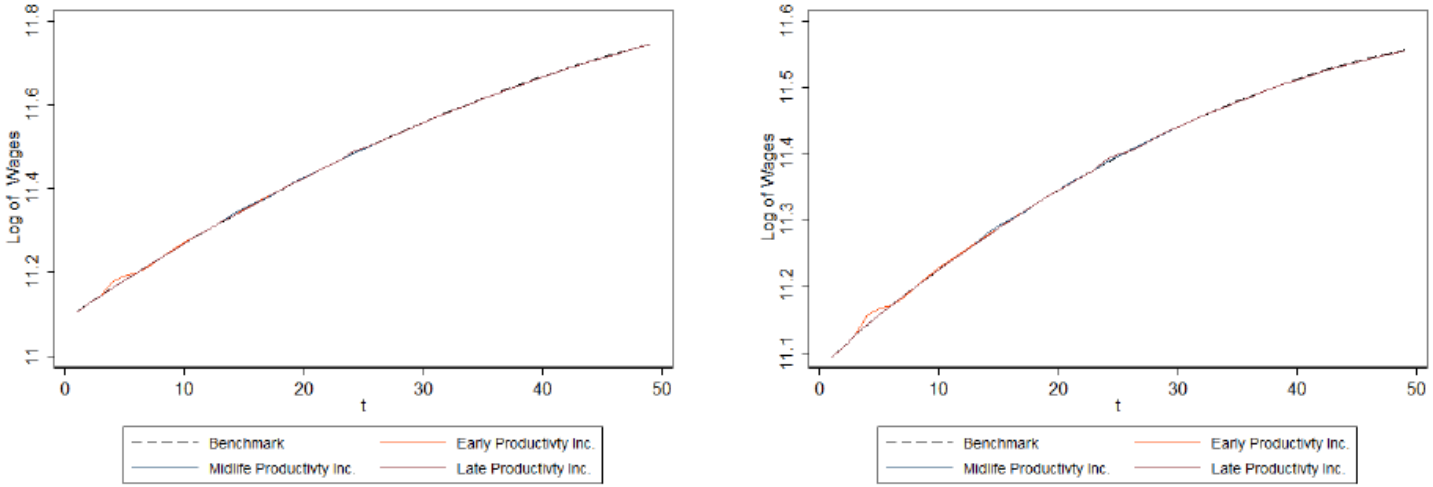


Women

Second Step: Simulations on Lifetime Earnings

Using our "representative" researcher, we simulated wage profiles,

Figure 3: Simulated Earnings Life-Cycle Profile for Different Productivity Changes



Men *Women*

Second Step: Simulations on Lifetime Earnings

Using our "representative" researcher, we simulated wage profiles, and computed

$$w^s = \sum_{t=1}^{50} \frac{s_t}{(1+r)^t}$$

assume that $r = 0.03$ as Divide and Koenker (1991).

Table 4: Simulation of Lifetime Earnings of Publishing and Additional 3 Pages Paper

	Benchmark		*Early Productivity Change*		*Mid-life Productivity Change*		*Late Productivity Change*	
	Men	Women	Men	Women	Men	Women	Men	Women
PV Lifetime Earnings	2,300,786.3	2,111,674.5	2,302,022.9	2,112,977.4	2,300,510.7	2,111,491.5	2,300,205.5	2,111,196.6
Annuity Value	46,955.82	43,095.04	46,980.99	43,122.18	46,949.66	43,092.95	46,943	43,086.62
Difference								
PV Lifetime Earnings			1,235.60	1,302.90	-276.60	-183.00	-581.80	-478.90
Annuity Value			25.22	26.59	-5.64	-3.73	-11.87	-9.77

Thanks